

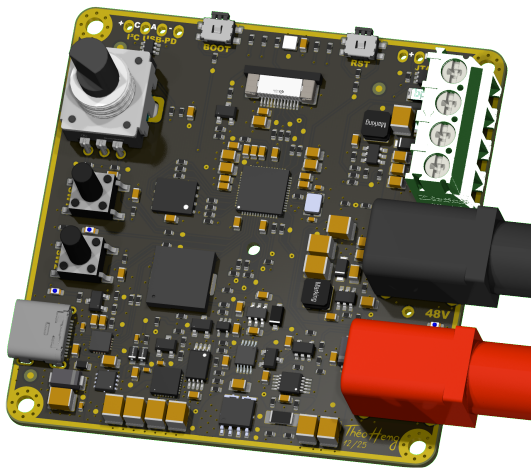
PD240W

Variant: RELEASED

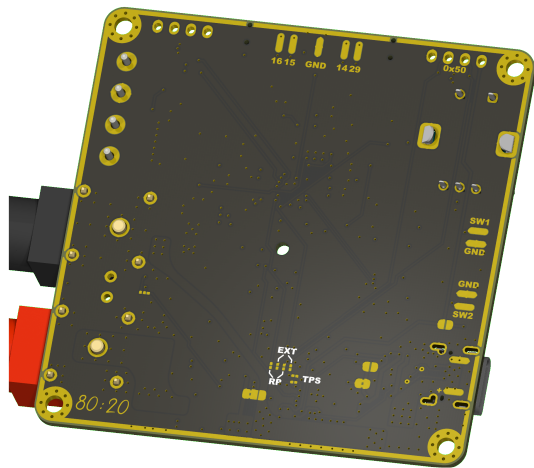
2026-03-15
Rev 1.0.0

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	Power - Sequencing	21		31	
2	Block Diagram	12	Revision History	22		32	
3	Project Architecture	13		23		33	
4	Power - USB-C Input	14		24		34	
5	Power - PD Controller	15		25		35	
6	Power - Sensing	16		26		36	
7	Power - Output	17		27		37	
8	MCU - Unit	18		28		38	
9	User - Interface	19		29		39	
10	Power - Generation	20		30		40	

TOP VIEW



BOTTOM VIEW



NOTES

An adjustable power supply using USB-C Power Delivery negotiation, supporting up to 240W at 48V 5A. This device is designed to be compatible with USB-PD 3.1 and above.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 15-Mar-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

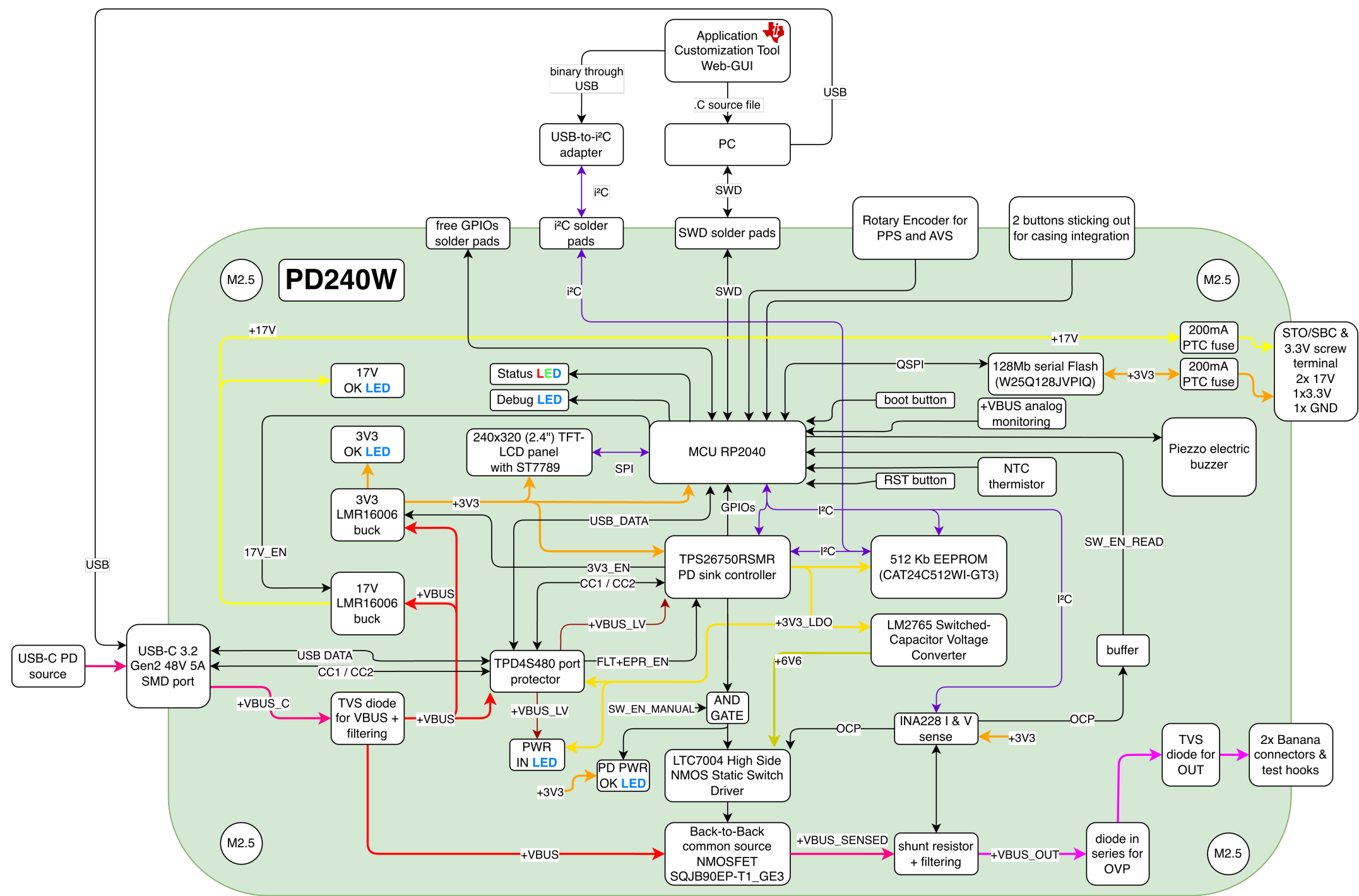
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ac7b12e	
	Board Name: PD240W		Project Name: PD240W		
Sheet Title:	File Name: PD240W_kicad.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0	
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 12	

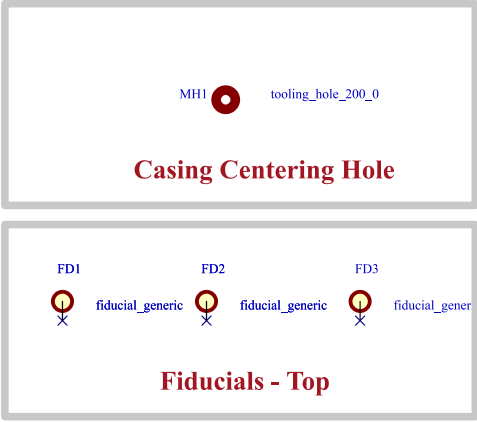
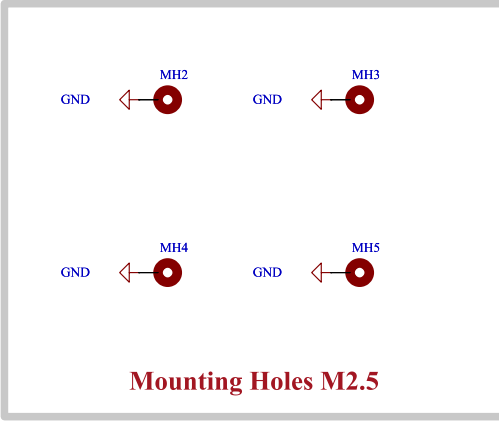
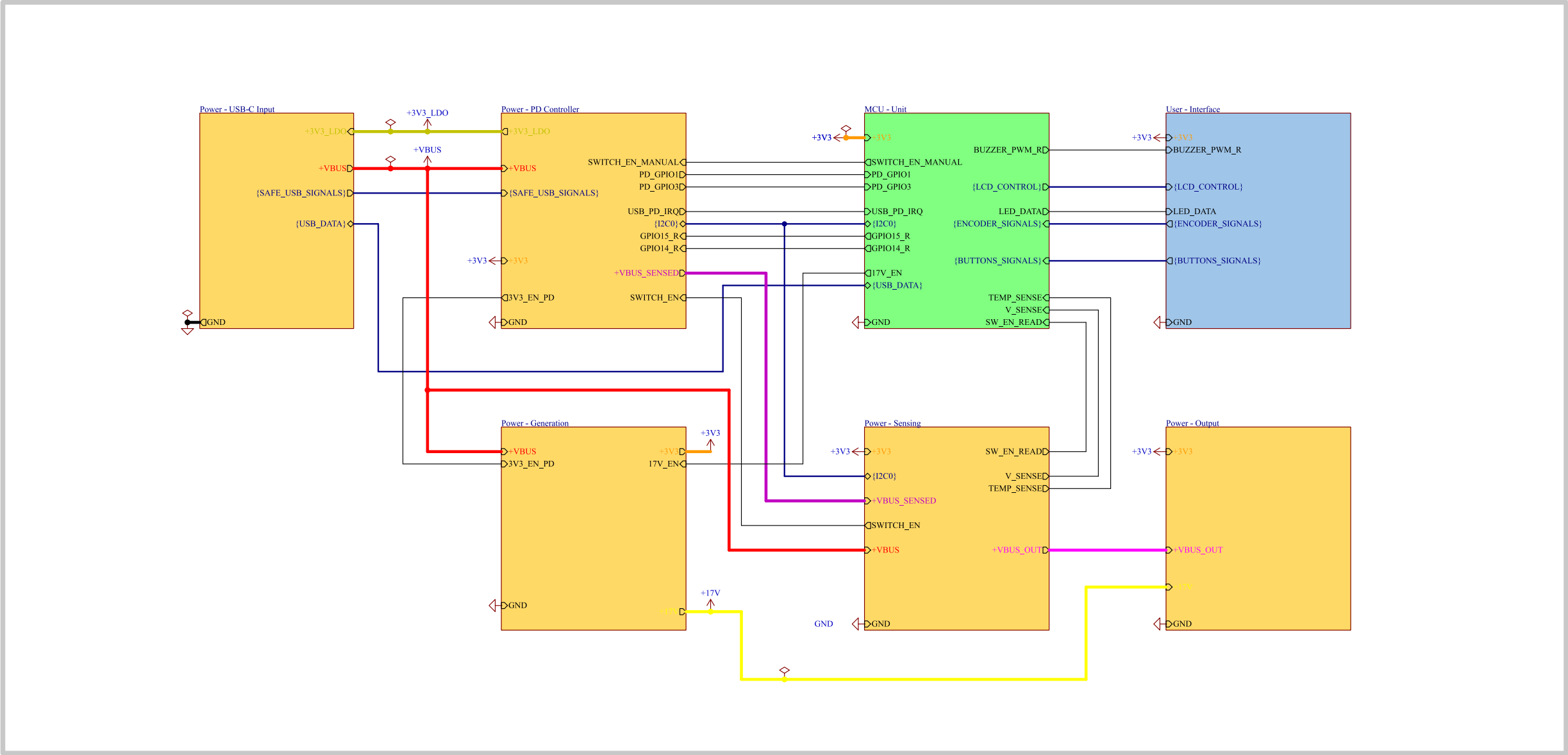
[2] Block Diagram



Target specifications:	
Input voltage:	5 - 53 V
Output Voltage:	3.3 - 48 V
Max. Output Current:	5 A
Max Power Output:	240 W

Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ac7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 12

[3] Project Architecture



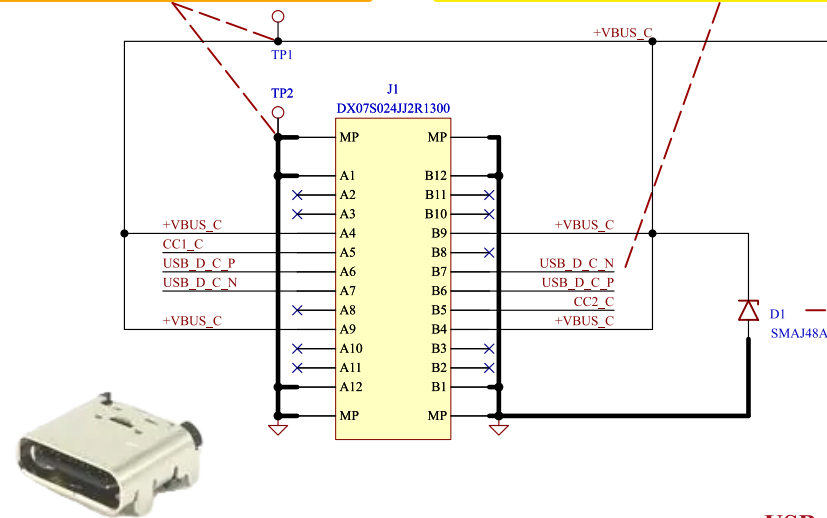
Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ac7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 12

[4] Power - USB-C Input

DEBUG NOTE:
+VBUS_C and GND solder pads to test PCB
with external supply at limited current.

DESIGN NOTE:
Pin A6 & B6 as well as A7 & B7 need to be connected together,
to be able to use the USB-C connector in both directions.

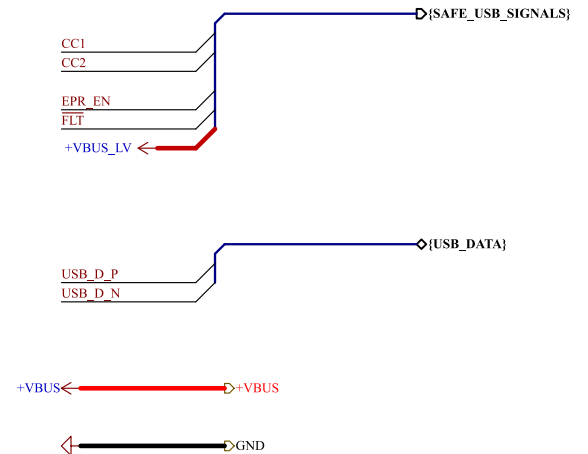
LAYOUT NOTE:
Place the capacitors next to the VBUS pins.



DESIGN NOTE:
 $V_r = 48V$
 $V_{brmin} = 53.3V$
 $V_c = 77.4V$
This diode is our best option for 48V, but it won't be enough as TPD4S480 has a 63V maximum input voltage, and LMR16006 65V.

DESIGN NOTE:
22ohm @ 100Mhz
4mohm @ DC
8A rated @ 85°C

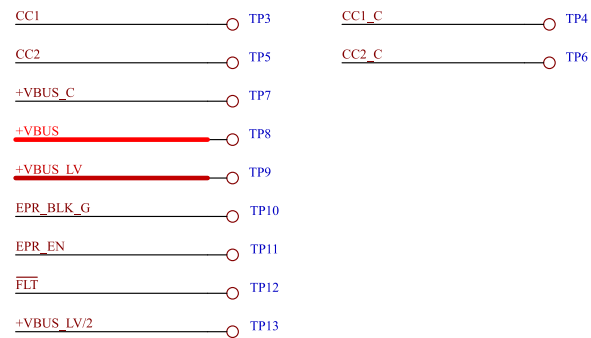
USB-C 3.2 Gen2 48V 5A SMD receptacle



Output Signals

+3V3_LDO → +3V3_LDO

Input Signals



Test Points

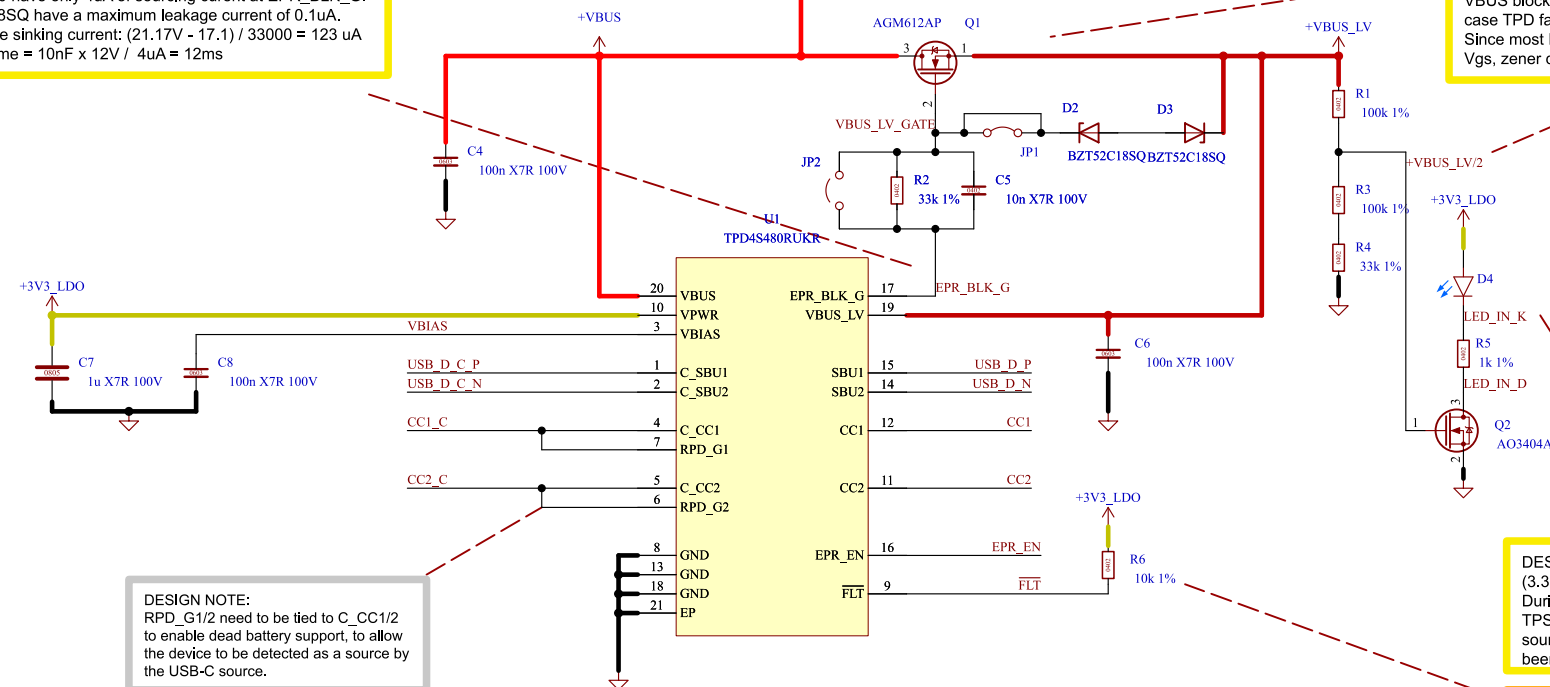
DESIGN NOTE:
TPD4S480 have only 4uA of sourcing current at EPR_BLK_G.
BZT52C18SQ have a maximum leakage current of 0.1uA.
Worst case sinking current: $(21.17V - 17.1) / 33000 = 123 \mu A$
Turn-on time = $10nF \times 12V / 4\mu A = 12ms$

DESIGN NOTE:
VBUS blocking fet needs to have $V_{ds} \geq 48V$ (in case TPD fails) and $V_{gs} \geq 0.42 \times 50.4V = 21.16V$.
Since most FETs in this package have a $\pm 20V$ V_{gs} , zener diodes are used to clamp V_{gs} to 18V.

DESIGN NOTE:
 $+VBUS_{LV/2} = 2.85V @ +VBUS_{LV} = 5V$ (min)
 $14.3V @ +VBUS_{LV/2} = 25V$ (max)
 $V_{gs} < +VBUS_{LV/2} < V_{gs} \text{ max}$
 $2.6V < +VBUS_{LV/2} < 20V$

DESIGN NOTE:
 $(3.3 - 2.65) / 1000 = 0.65mA$
During dead-battery boot mode,
TPS26750 internal LDO's can only source 5mA, this is why LED current as been reduced.

DEBUG NOTE:
FLT is open-drain and requires a pull-up. To save board space, it could be removed, as TPS26750 has configurable internal pull-ups.



DESIGN NOTE:
RPD_G1/2 need to be tied to C_CC1/2 to enable dead battery support, to allow the device to be detected as a source by the USB-C source.

USB Type-C™ 48V EPR port protector with short-to-VBUS overvoltage and IEC ESD protection

Comments:	Company:		Variant:	Git Hash:
			RELEASED	ac7b12c
Sheet Title:	Board Name:		Project Name:	
	Pb240W		PD240W	
Power - USB-C Input	File Name:	Designer:	Date:	Revision:
	power_usb_c_input.kicad_sch	Theo Heng	2026-02-27	1.0.0
Sheet Path:	Reviewer:		Size:	Sheet:
/Project Architecture/Power - USB-C Input/			A3	4 of 12

[5] Power - PD Controller

DESIGN NOTE:
The default state for a USB-C port (before or during negotiation, or if negotiation fails) is effectively USB 2.0 specs: 5V @ 500mA. Therefore we want to limit the Inrush current. Synapticon drives have an average of 150uF of Input capacitance. As explained in application Informations of LTC7004, the gate capacitor and TGUP resistors are chosen to limit the current to a safe limit of around 250mA.

$I_{inrush} = (0.7 \times 12V \times 150uF) / (50k\Omega \times 100nF) = 252mA$

Time to reach 48V = $(150uF \times 48) / 0.252A = 29ms$

Boost capacitor adjustments: SQJB90EP has a max gate charge of 2x25nC = 50nC
The relationship for CG that needs to be maintained when CG is used is given by: $C_b > (mosfet\ Q_g) / 1V + 10 \times CG = 1.05\ uF$
With derating, a 2.2uF cap is necessary.

DESIGN NOTE:
3V3 LDO total current needs to be lower than 5mA at startup:
EEPROM read: 1mA
POWER OK LED: 0.65mA
TPD4S480: 0.16mA
LM2765: 0.13mA
LTC7004: 0.05mA
I²C pull-ups: ~0.5mA
ADC resistors: 0.02mA
AND gate: <0.01mA
Total ~2.5mA < 5mA

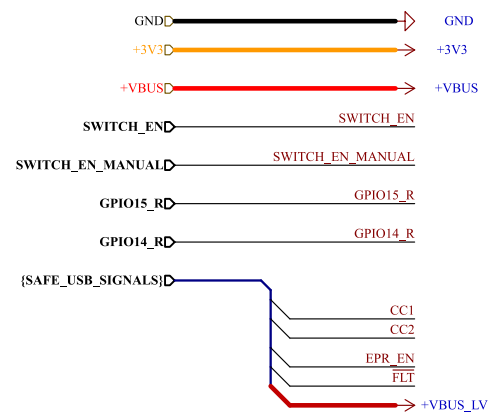
DEBUG NOTE:
If the gate driver is not working as expected, connect the bootstrap capacitor to +VBUS instead of TS.

DEBUG NOTE:
ADCIN both connected to LDO1.5V result in ADCINx decoded value 5, which activate AlwaysEnableSink mode and I²C address 0x21 (7 bits). (see page 34)

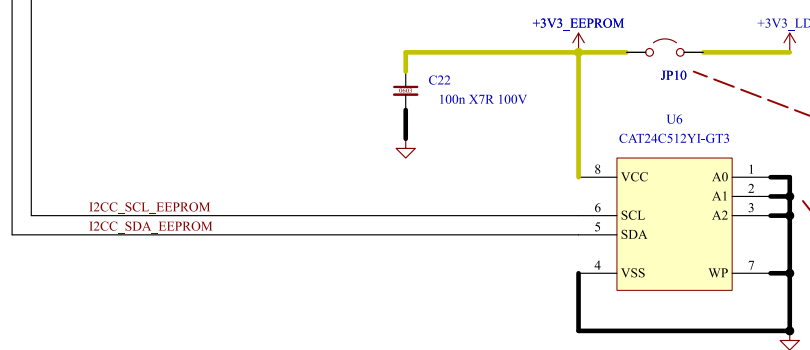
DESIGN NOTE:
LED current:
 $(3.3 - 2.65) / 330 = 2mA$

USB Type-C® and USB PD Controller With Integrated Power Switches

LAYOUT NOTE:
CC1/CC2 capacitors need to be placed as close as possible to their respective pins and on the same layer as the TPS26750 device.



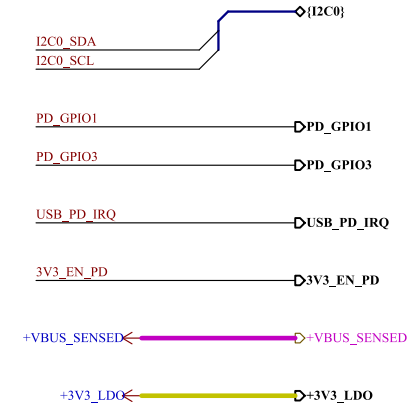
Input Signals



512 Kb I2C CMOS Serial EEPROM

DEBUG NOTE:
When flashing the binary configuration to the EEPROM leave the jumper open, and power the EEPROM from an external supply, to avoid I²C bus conflict with TPS26750.

DEBUG NOTE:
EEPROM Address: Must be 0x50 (7-bit) on the I2C bus for the TPS26750 to find it automatically. (1010000)



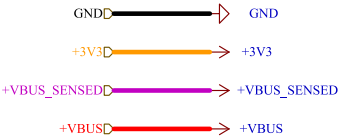
Output Signals



Test Points

Comments:	Company:	Variant:	Git Hash:
		RELEASED	ac7b12c
Sheet Title:	File Name:	Designer:	Date:
Power - PD Controller	power_pd_controller.kicad_sch	Theo Heng	2026-02-27
Sheet Path:	Reviewer:	Size:	Sheet:
/Project Architecture/Power - PD Controller/		A3	5 of 12

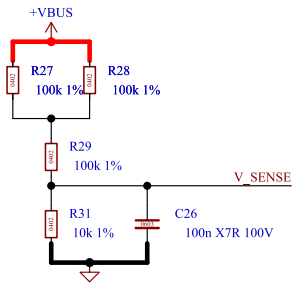
[6] Power - Sensing



Input Signals

DESIGN NOTE:
Network ratio = 1/16 = 0.0625
@48V V_SENSE = 3V
@5V V_SENSE = 0.31V

LAYOUT NOTES:
Place capacitor close to MCU DC pin. Route away from noisy traces.

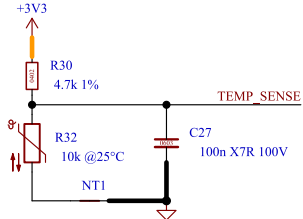


Input Voltage Sensing

DESIGN NOTE:
B-Constant is 3950K
1.2k @ 80°C -> 0.67V
2.5k @ 60°C -> 1.14V
10k @ 25°C -> 2.24V
13k @ 20°C -> 2.42V

LAYOUT NOTES:
NTC must be placed close and thermally coupled to the ground plane near the relevant heat source. Route away from noisy traces.

DEBUG NOTE:
Low side measurement is used to improve thermal coupling, which invert the maths in software.

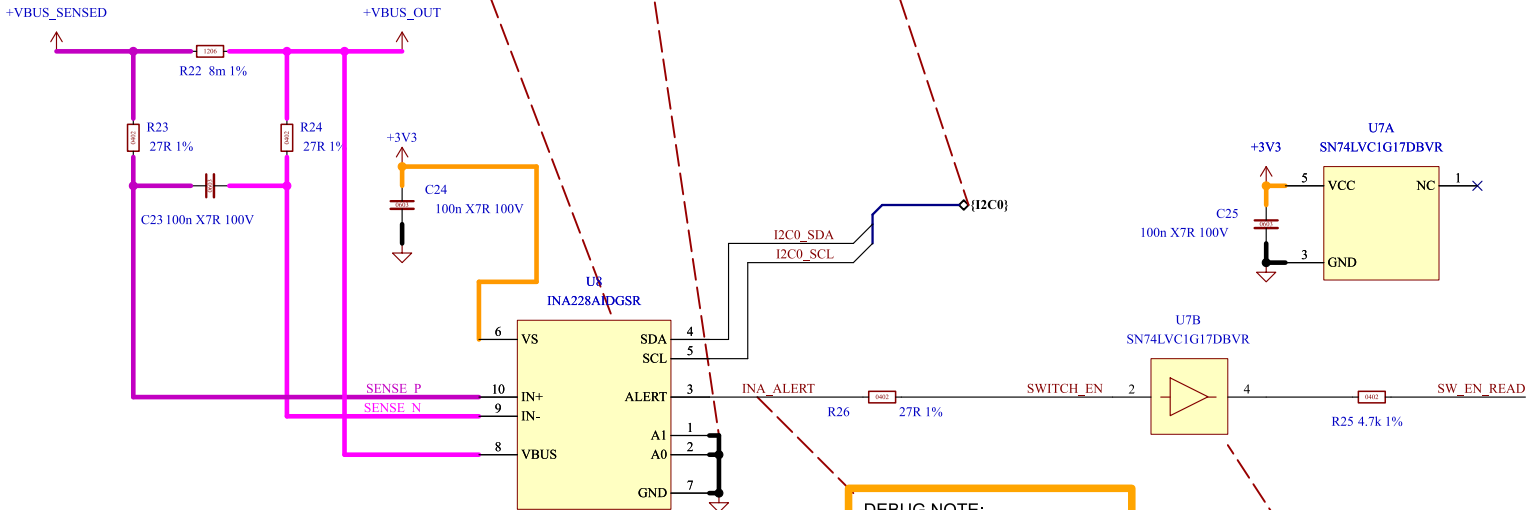


Temperature Sensing

DESIGN NOTE:
ADCRANGE = 1 (+/- 40.96 mV range)
 $V_{SENSE} = 5A \times 0.008 = 40\text{ mV}$. This utilizes 97.6% of the ADC range, providing the highest possible resolution.
Power Dissipation: $P = 5^2 \times 0.008 = 0.2W$

DEBUG NOTE:
INA228AIDGSR address when A1 & A0 are tied to GND is 0x40 (7 bits 1000000) (see page 19)

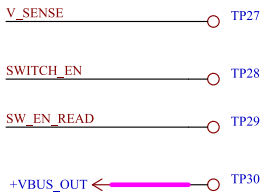
DESIGN NOTE:
I2C pullups are on Power - PD Controller schematic



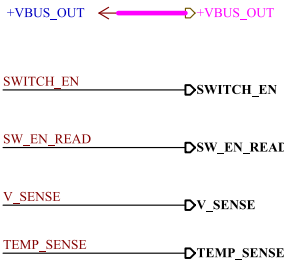
85-V, 20-Bit, Ultra-Precise Power/Energy/Charge Monitor With I²C Interface

DEBUG NOTE:
ALERT is open-drain and can be configured to latch or not latch the overcurrent event.

DESIGN NOTE:
SN74LVC1G17 is a safety buffer to prevent software mistakes. Driving SWEN_READ cannot activate the power switch.



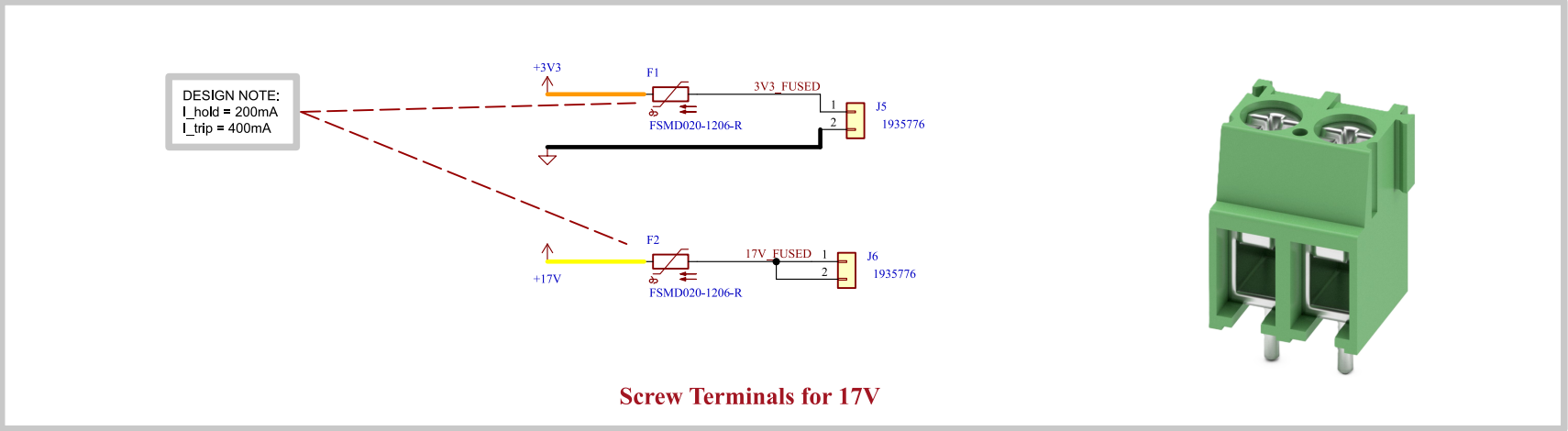
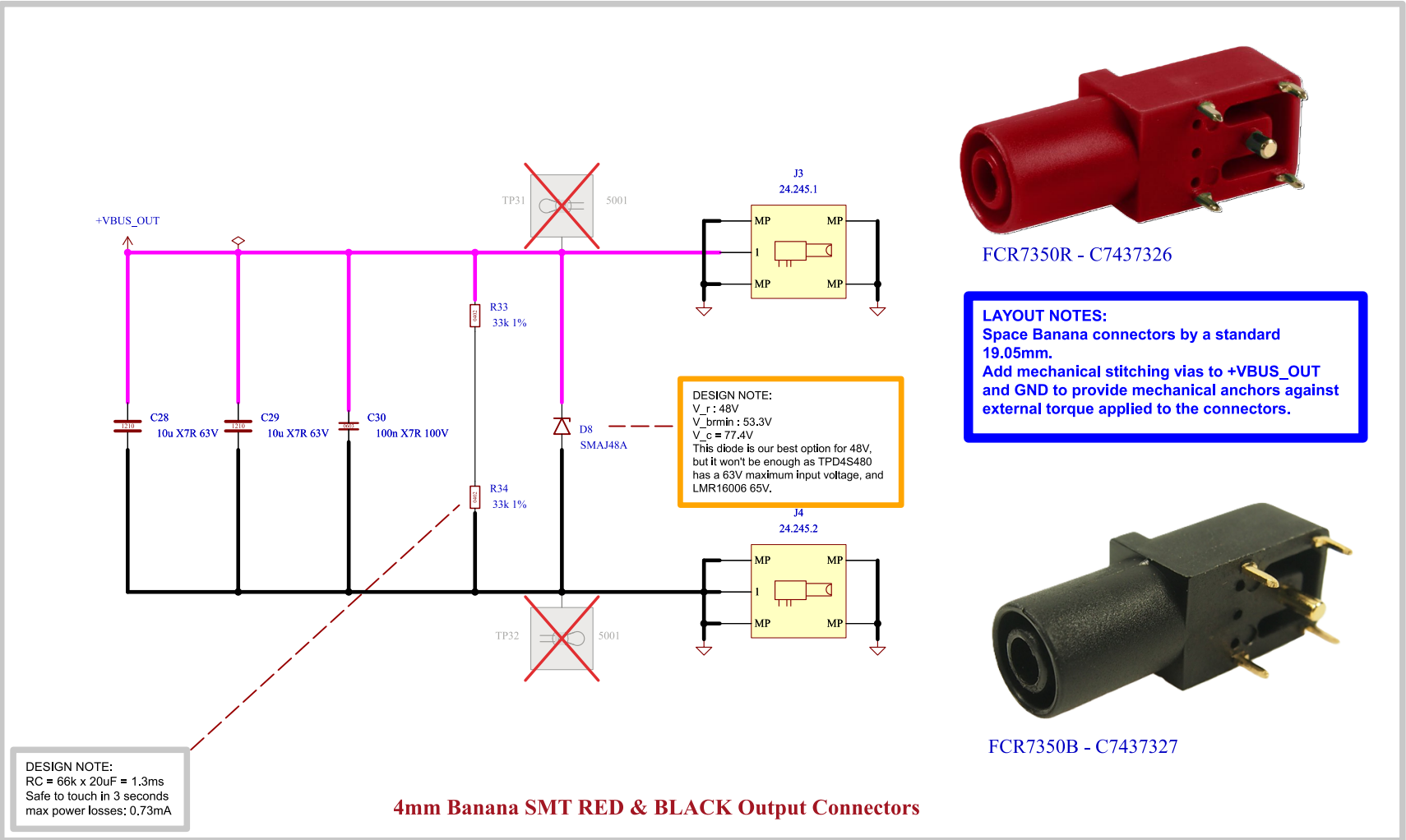
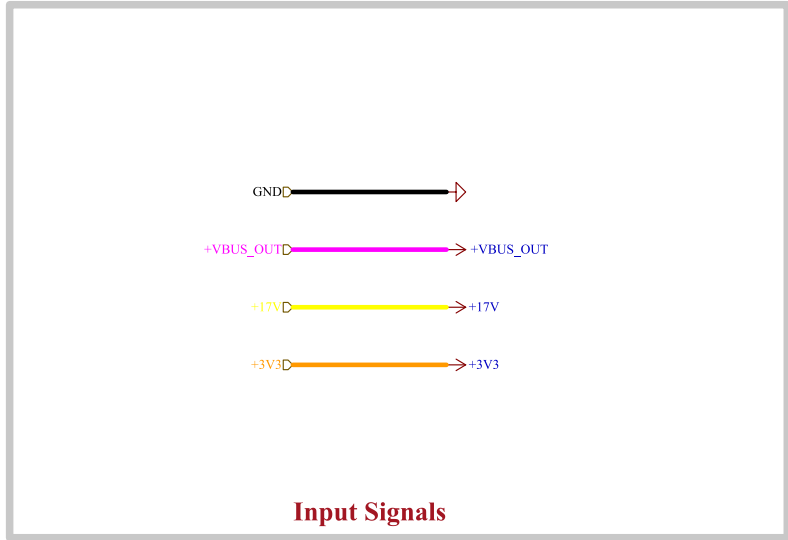
Test Points



Output Signals

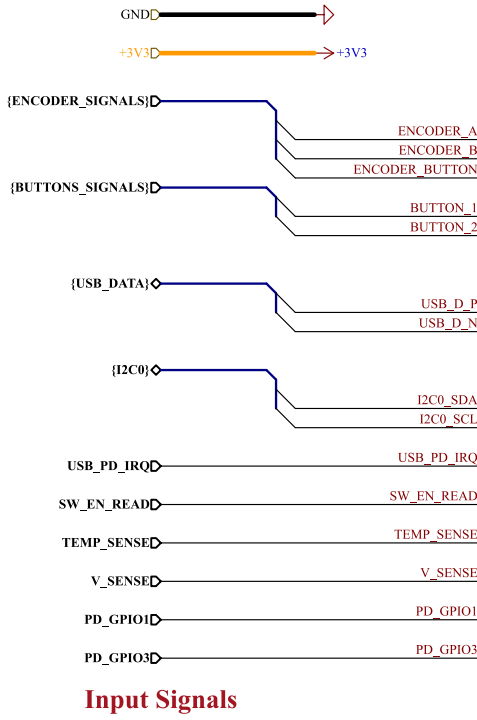
Comments:	Company:		Variant:	Git Hash:
			RELEASED	ac7b12c
Sheet Title:	Board Name:		Project Name:	
	Power - Sensing		PD240W	
Sheet Path:	File Name:	Designer:	Date:	Revision:
	/Project Architecture/Power - Sensing/	Theo Heng	2026-02-27	1.0.0
			Size:	Sheet:
			A3	6 of 12

[7] Power - Output

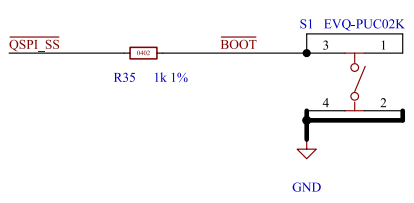


Comments:	Company:		Variant:	Git Hash:
			RELEASED	ac7b12e
Sheet Title:	Board Name:		Project Name:	
	Power - Output		PD240W	
Sheet Path:	File Name:	Designer:	Date:	Revision:
	/Project Architecture/Power - Output/	Theo Heng	2026-02-27	1.0.0
			Size:	Sheet:
			A3	7 of 12

[8] MCU - Unit

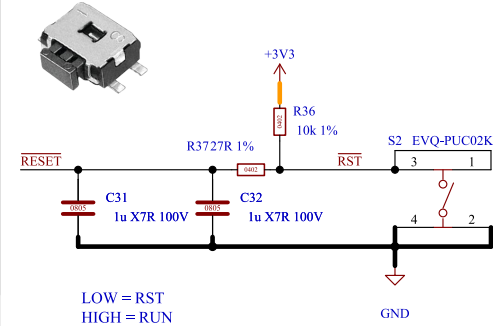


Input Signals



MCU Boot Button

LAYOUT NOTES:
D+ and D- lines of USB-C needs to be routed as differential pairs, with 90Ω impedance.
Spacing: 0.2032mm
Width: 0.15mm



MCU Reset Button

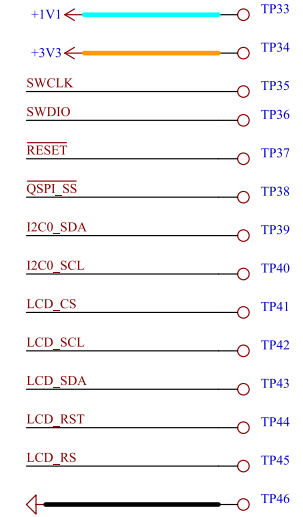
DESIGN NOTE:
We want to delay the start of the MCU to allow power to stabilize and avoid flash corruption. RP2040 is assured to be turned on when the RUN pin reaches 2.0V, but can turn on at 0.8V already. Run has an internal pull-up of 50k-80k. ->Effective pull-up is between 8.3k and 8.8k

RCmin = 8.3k x 2uF = 16.6ms
RCmax = 8.8k x 2uF = 17.6ms

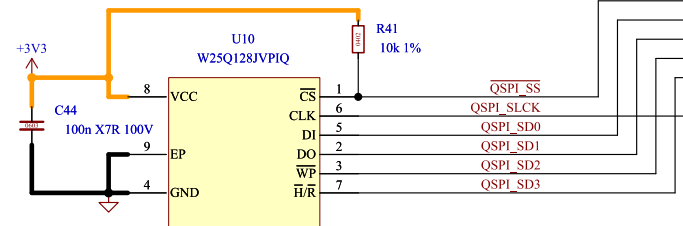
Worst case (fastest turn-on):
 $V(t) = V \times (1 - e^{-(t/RC)})$
 $0.8 = 3.3 \times (1 - e^{-(t/RC)})$
 $-t/RC = \ln(0.757)$
 $t = 0.28 \times RCmin = 4.6ms$

Worst case (slowest turn-on):
 $2 = 3.3 \times (1 - e^{-(t/RC)})$
 $-t/RC = \ln(0.394)$
 $t = 0.931 \times RCmax = 16.38ms$

Has been verified in Falstad simulation.

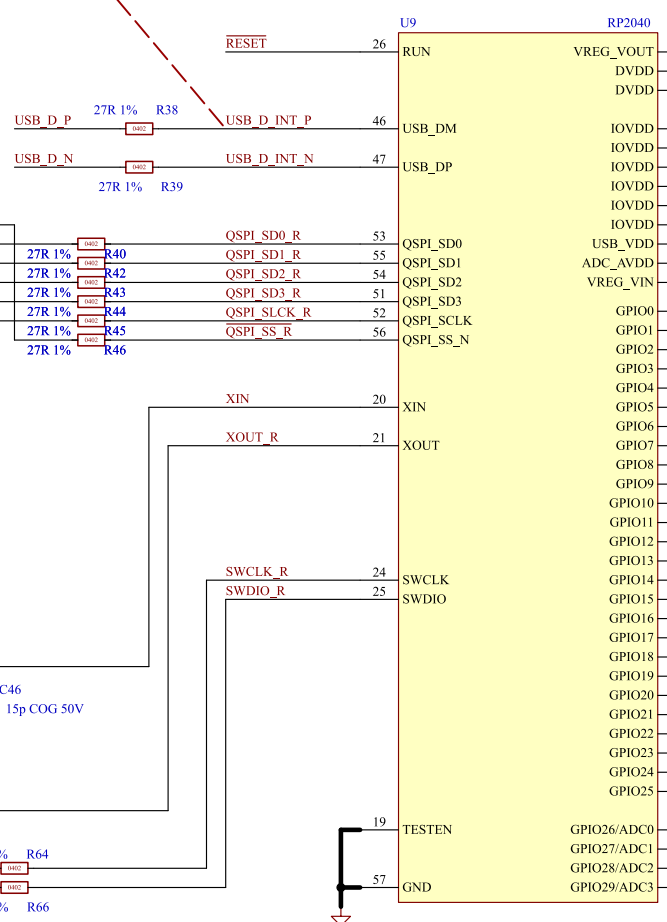
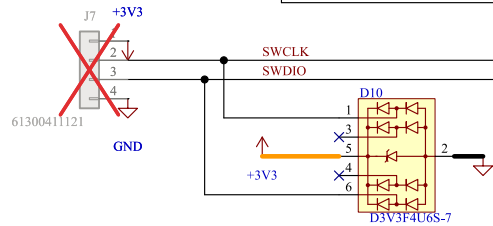
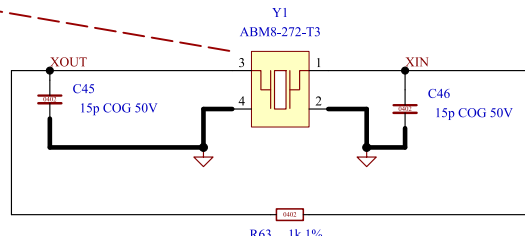


Test Points

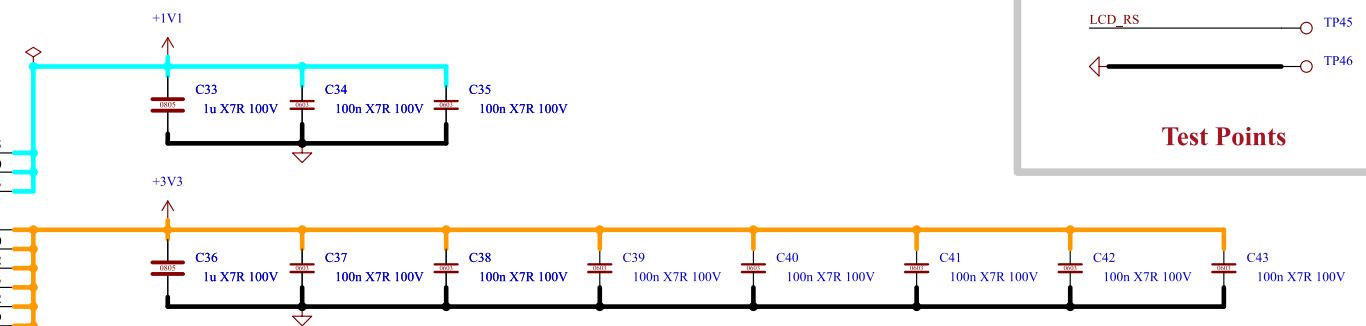


128Mb Serial Flash Memory

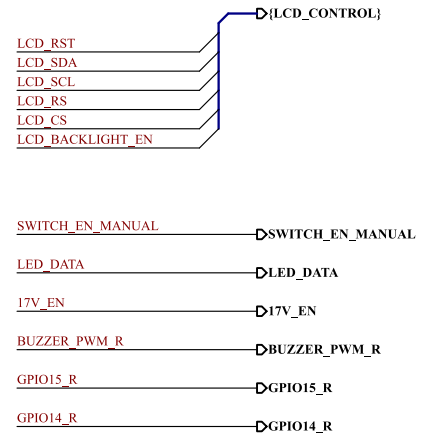
DESIGN NOTE:
Recommended crystal is ABM8-272-T3 at 12Mhz.



MCU - Unit



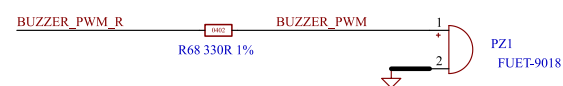
DESIGN NOTE:
LED current: $(3.3 - 2.65) / 330 = 2mA$



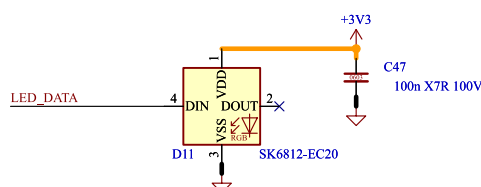
Output Signals

Comments:	Company:		Variant:	Git Hash:
	RELEASED		RELEASED	ae7b12c
Sheet Title:	Board Name:		Project Name:	
	MCU - Unit		PD240W	
Sheet Path:	File Name:	Designer:	Date:	Revision:
	/Project Architecture/MCU - Unit/	Theo Heng	2026-02-27	1.0.0
			Size:	Sheet:
			A3	8 of 12

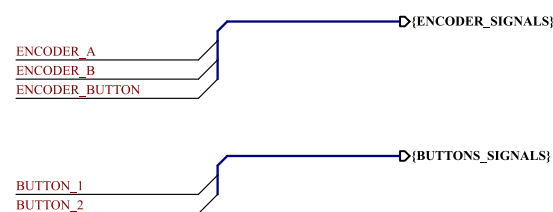
[9] User - Interface



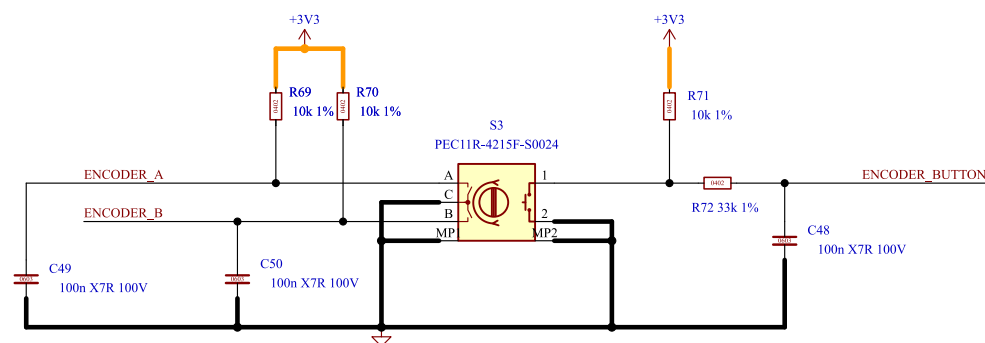
9x9mm SMD 65dB piezoelectric buzzer



Addressable RGD LED



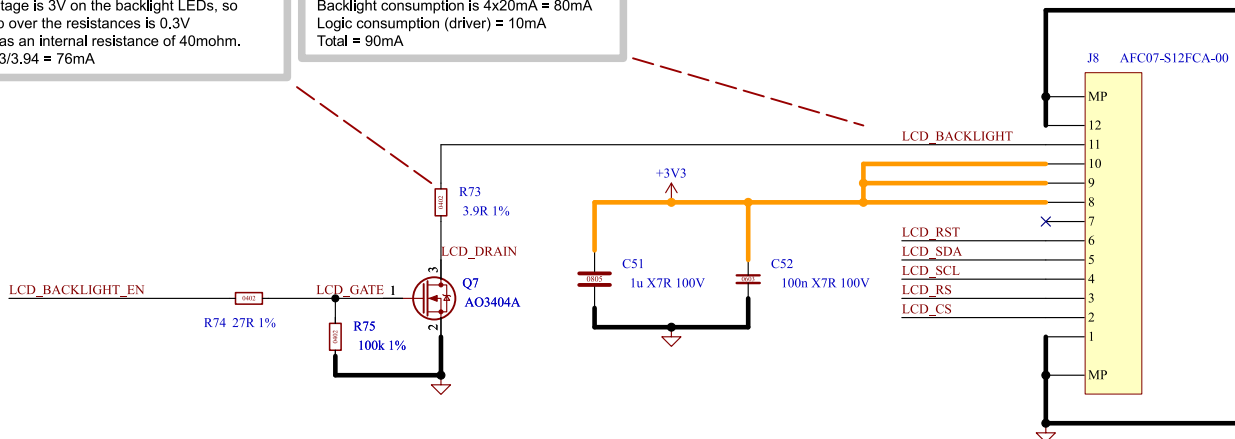
Output Signals



24 PPR Rotary Encoder with 24 Detents & Button

DESIGN NOTE:
Forward voltage is 3V on the backlight LEDs, so voltage drop over the resistances is 0.3V
AO3404A has an internal resistance of 40mohm.
current = $0.3/3.94 = 76\text{mA}$

DESIGN NOTE:
Backlight consumption is $4 \times 20\text{mA} = 80\text{mA}$
Logic consumption (driver) = 10mA
Total = 90mA



12-pins, Low-side Contact FPC Connector and Backlight Driver

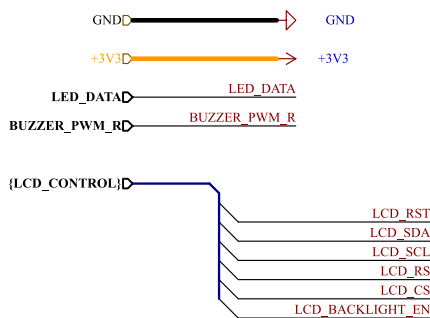
LAYOUT NOTES:
FPC connector of LCD display is 20.7mm long, which allows for around 10mm of clearance between the top of the PCB and the bottom of the display.



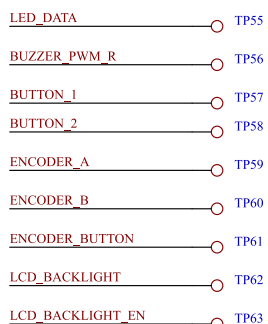
NO.	SYMBOL
1	GND
2	CS
3	RS
4	SCL
5	SDA
6	RESET
7	NC
8	IOVCC
9	VCC
10	LEDA
11	LEDK
12	GND



240x320 (2.4") TFT-LCD Panel with ST7789 driver & FPC

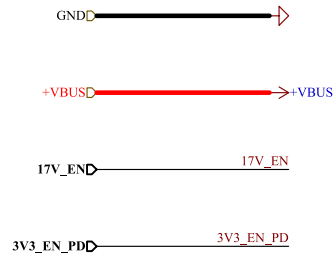


Input Signals

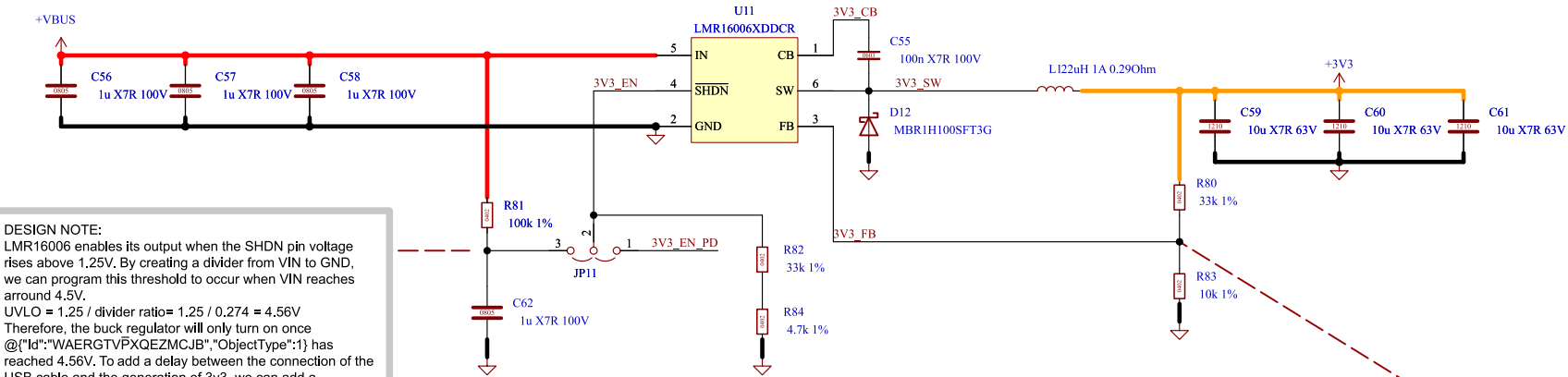


Test Points

Comments:	Company:		Variant:	Git Hash:
			RELEASED	ac7b12e
Sheet Title:	Board Name:		Project Name:	
	User - Interface		PD240W	
Sheet Path:	File Name:	Designer:	Date:	Revision:
	user_interface.kicad_sch	Theo Heng	2026-02-27	1.0.0
/Project Architecture/User - Interface/		Reviewer:	Size:	Sheet:
			A3	9 of 12



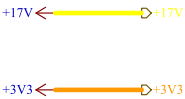
Input Signals



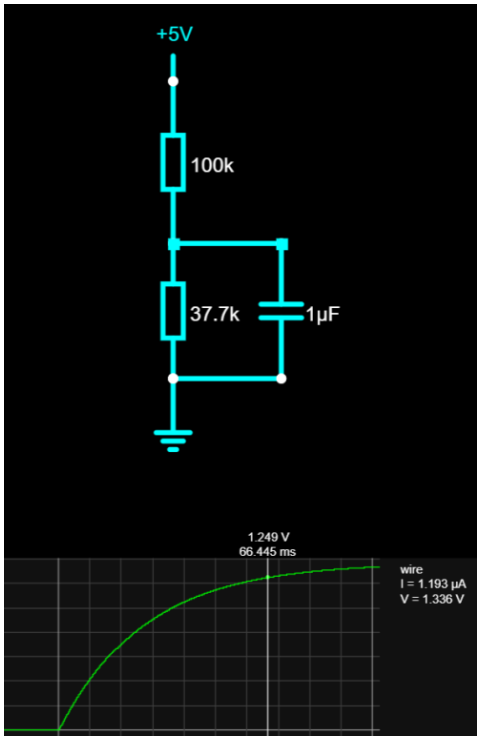
DESIGN NOTE:
LMR16006 enables its output when the SHDN pin voltage rises above 1.25V. By creating a divider from VIN to GND, we can program this threshold to occur when VIN reaches around 4.5V.
 $UVLO = 1.25 / \text{divider ratio} = 1.25 / 0.274 = 4.56V$
Therefore, the buck regulator will only turn on once @ ("Id": "WAERGTVPXQEZMCJB", "ObjectType": 1) has reached 4.56V. To add a delay between the connection of the USB cable and the generation of 3v3, we can add a capacitor.
From Thevenin equivalent resistance:
 $RC = 27k \times 1\mu F = 27ms$
 $V(t) = V_{final} \times (1 - e^{-(t/RC)})$
 $1.25 = 1.37 \times (1 - e^{-(t/RC)})$
 $-t/RC = \ln(0.087)$
 $t = 2.44 \times RC = 66ms$
LMR16006 should turn on 66ms after VBUS has reached 5V.

3.3V 600mA 700kHz Buck Regulator

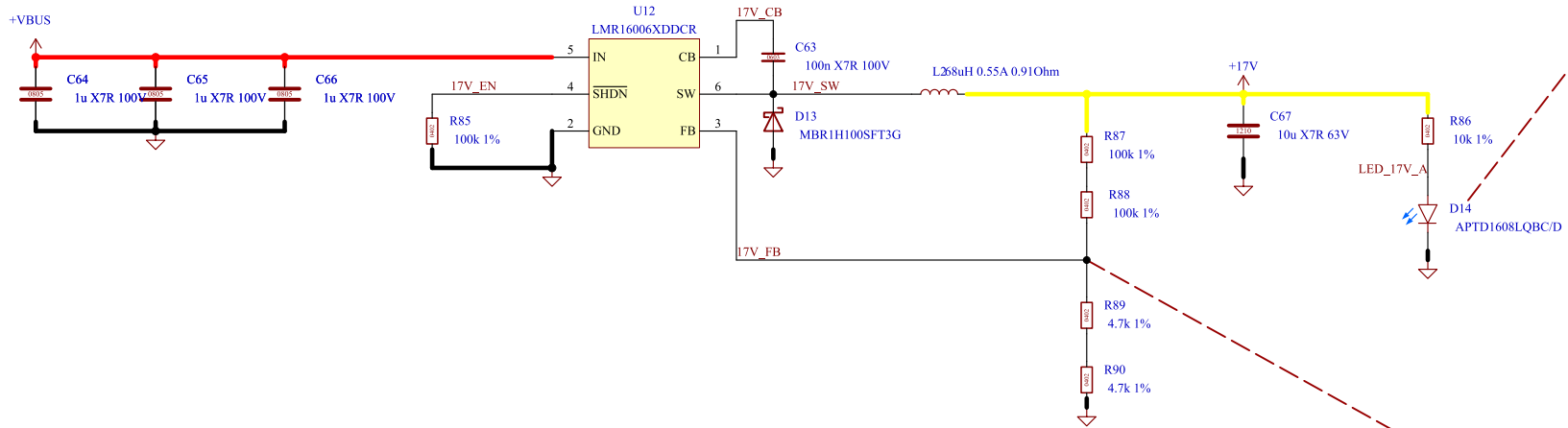
DESIGN NOTE:
FB: $V_{out} = 0.765 \times (1 + R_{top}/R_{bot}) = 0.765 \times (1 + (33k / 10k)) = 2.29V$
Optimized for 600mA:
Inductor: $(48 - 3.3) \times 3.3 / (700000 \times 48 \times 0.18) = 24\mu H$
22uH was selected as a close standard value.



Output Signals



Falstad Simulation Results: 1.25V reached in 66ms.



DESIGN NOTE:
LED current: $(17 - 2.65) / 10000 = 1.4mA$

17V 600mA 700kHz Buck Regulator

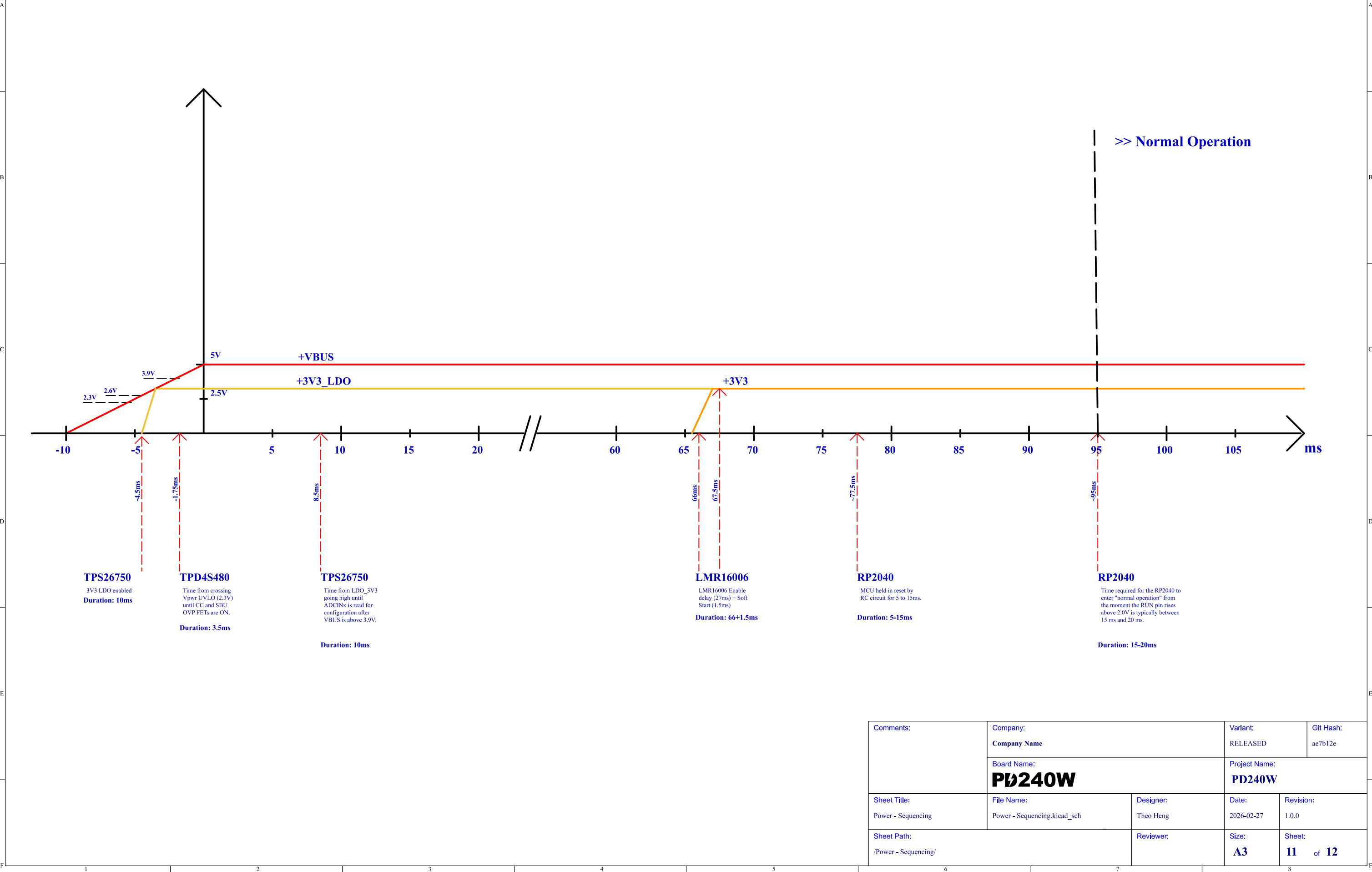
DESIGN NOTE:
FB: $V_{out} = 0.765 \times (1 + R_{top}/R_{bot}) = 0.765 \times (1 + (200k / 9.4k)) = 17.04V$
Optimized for 600mA:
Inductor: $(48 - 17) \times 17 / (700000 \times 48 \times 0.18) = 87\mu H$
68uH was selected as a close standard value.



Test Points

Comments:	Company:		Variant:	Git Hash:
			RELEASED	ac7b12c
Sheet Title:	Board Name:		Project Name:	
	Power - Generation		PD240W	
Sheet Path:	File Name:	Designer:	Date:	Revision:
	/Project Architecture/Power - Generation/	Theo Heng	2026-02-27	1.0.0
			Size:	Sheet:
			A3	10 of 12

[11] Power - Sequencing



Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ac7b12c
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A3	Sheet: 11 of 12

[12] Revision History

Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ac7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 12 of 12